



KYAMBOGO UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF MECHATRONICS AND BIOMEDICAL ENGINEERING

COMPUTING II

GROUP E EVENING

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1. Introduction

This report describes the design and implementation of the **Hostel Room Allocation System** developed in the C programming language. The system is designed to automate the process of registering students, managing room allocations, and enforcing priority policies based on specific institutional rules.

2. Problem Statement and Objectives

Manual hostel allocation is prone to human error, double-booking, and inefficient room utilization. This project aims to deliver a console-based application that achieves the following objectives:

- **Student Registration:** Capture and store relevant student data.
- **Dynamic Search and Update:** Allow quick lookups and modifications of existing allocations.
- **Record Management:** Provide functionality to safely delete records.
- **Inventory Tracking:** Maintain and display real-time availability of rooms.
- **Rule Enforcement:** Automate allocation priority based on special needs, year of study, and room preferences.

3. System Design and Rules Implementation

3.1 Data Structures

The system uses custom structures (struct) to represent key entities cleanly:

- **Student:** Stores ID, Name, Year of Study, Special Needs Status, and Room Preference.
- **Room:** Stores Room Number, Type, Maximum Capacity, and Current Occupancy.

3.2 Rules Engine Logic

The program applies business logic to handle allocations automatically:

- **Capacity Constraint:** Before allocation, the program checks if `current_occupancy < max_capacity`.
- **Priority Queueing:** Sorting algorithms or conditional filters ensure students with special needs and final-year students are processed first when room availability is low.
- **Preference Matching:** The system matches the student's `room_type_preference` with available inventory.
- **Waiting List Management:** If all criteria fail and zero matching rooms are free, the system flags the student and triggers a waiting-list notification.

4. Program Modules and Functionality

Module / Function	Description
<code>registerStudent()</code>	Collects and validates new student records.
<code>searchAllocation()</code>	Searches records by Student ID or Room Number.
<code>updateAllocation()</code>	Modifies room assignments or student details.
<code>deleteAllocation()</code>	Frees up a room by removing an allocation record.
<code>displayAvailableRooms()</code>	Lists all rooms that are not yet at full capacity.
<code>generateAllocationReport()</code>	Outputs a complete breakdown, highlighting fully occupied rooms.

5. Testing and Sample Outputs

Test Cases

1. **Normal Allocation:** Allocating a standard student to their preferred room type.
2. **Priority Allocation:** Verifying a special-needs student bypasses standard applicants for a limited room.
3. **Waiting List Trigger:** Attempting to book a student when all corresponding rooms are at maximum capacity.

6. Conclusion

The Hostel Room Allocation System successfully optimizes student housing workflows. By embedding structural checks directly into the memory and logic flow, it eliminates double-booking errors and ensures that priority rules (such as special needs considerations) are handled fairly and transparently.